

Retrieval Augmented Generation

- ➔ The Need for RAG in your LLM-Powered Product
- ➔ RAG scenarios
- ➔ Vanilla RAG
- ➔ Embedding Vectors and Semantic Similarity
- ➔ Pros and Cons of Semantic Similarity - The Need for Reranking

The Need for RAG in your LLM-Powered Product

Knowledge cutoff limitations

- LLMs have training data cutoffs

- Can't access real-time or recent information without external retrieval

Privacy compliance

- Keeps sensitive data in controlled retrieval systems rather than baked into model parameters

Reduce the risk of hallucinations

- Provides source attribution

Domain-specific accuracy

- RAG enables access to specialized knowledge bases that general LLMs lack

Handle long-tail entities

- LLMs are known to not do well on items that are rare in the training data

- More likely to get right answers about Einstein as compared to a an assistant prof somewhere

RAG scenarios

Retrieving external knowledge

- Plug knowledge gaps of LLM

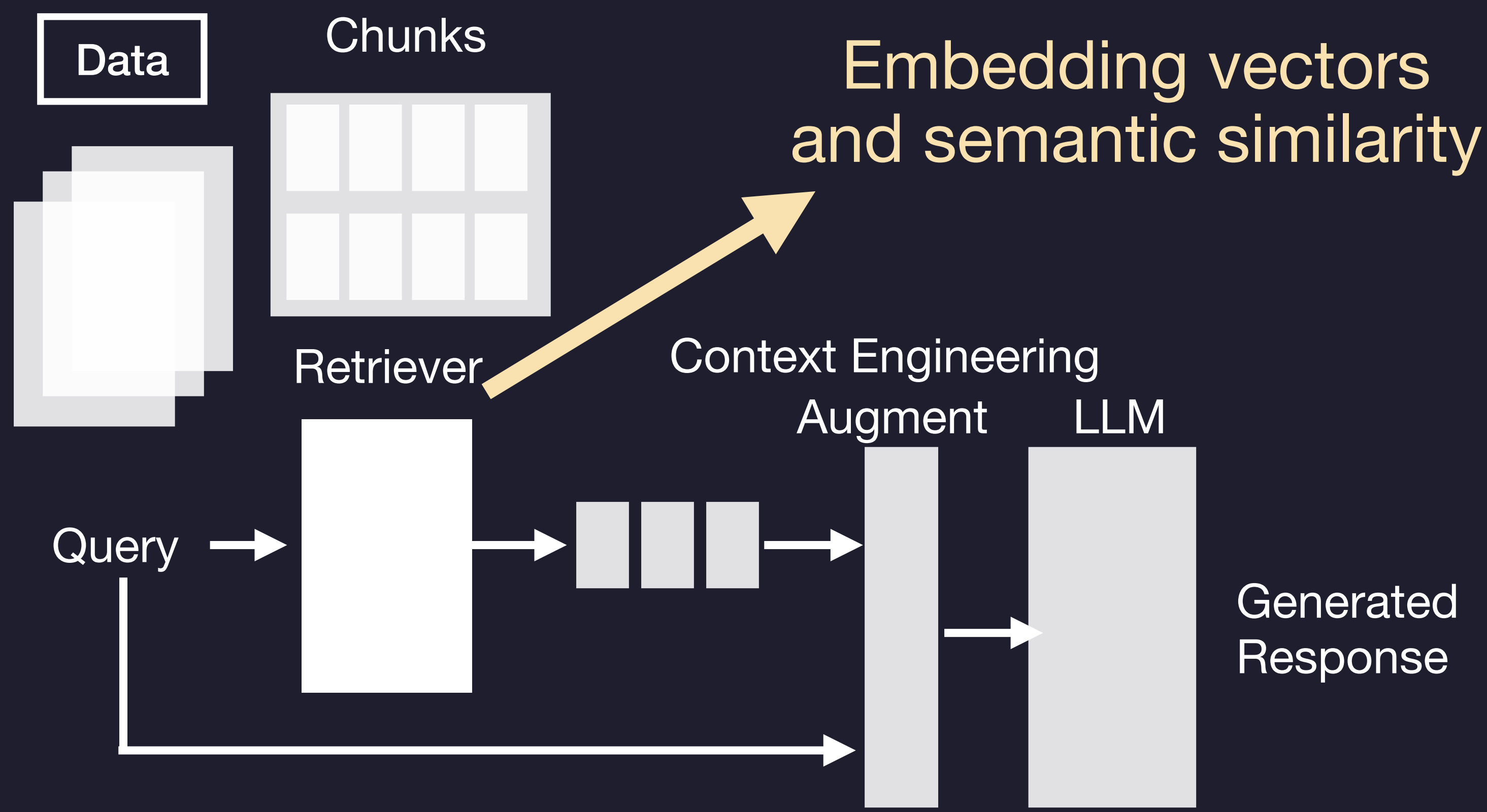
Retrieving context history

- Long term conversations

Retrieving in-context training examples

- Help LLMs get acquainted new task

Vanilla RAG



Chunks are semantically coherent parts of a document

LLMs have a fixed input window; chunking breaks long documents into manageable pieces that fit within this limit

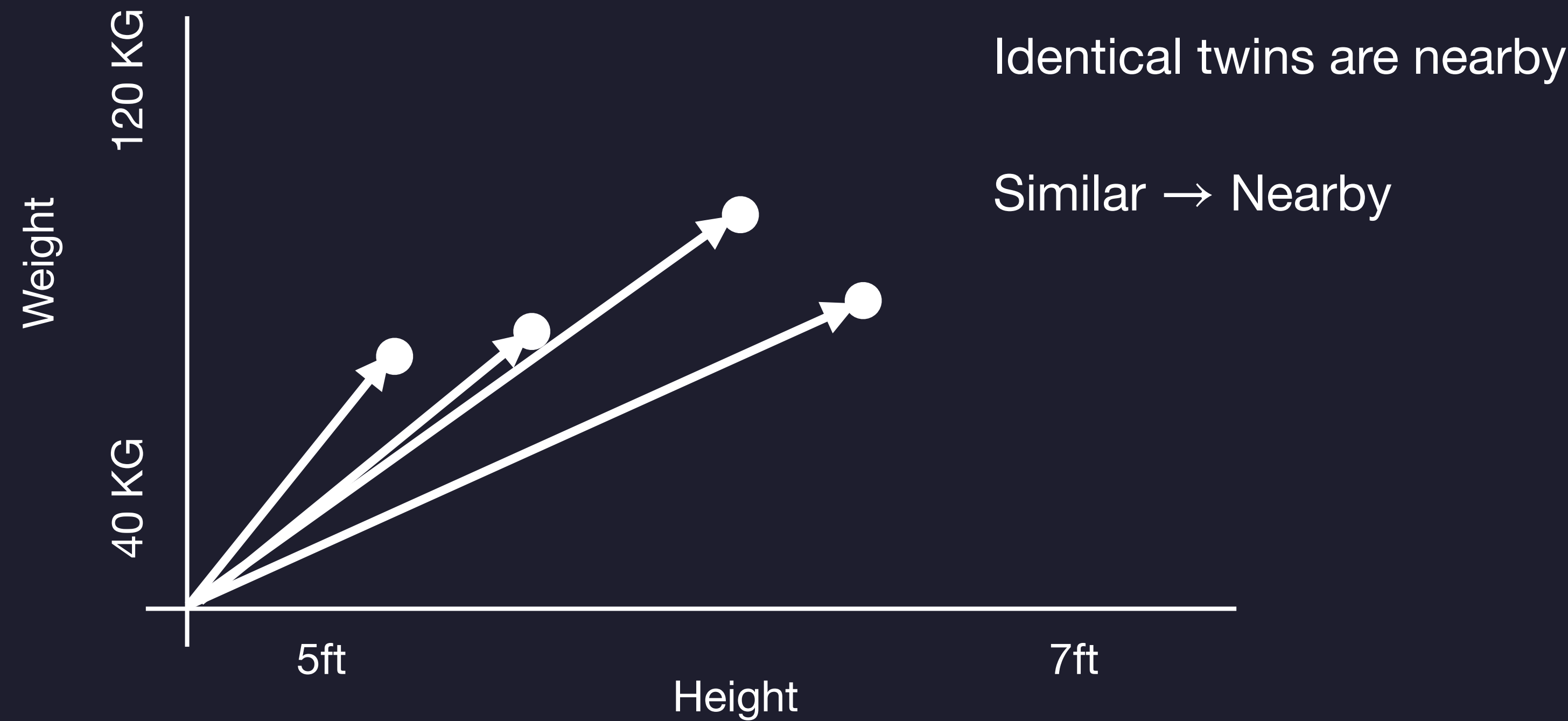
Chunking enables retrieval of only relevant information - avoids clutter in the context



A lot of engineering efforts go into optimizing chunking

Embedding Vectors and Semantic Similarity

Height	Weight
5ft, 5in	60 KG
5ft, 7in	65 KG
6ft, 2in	82 KG
5ft, 1in	48 KG



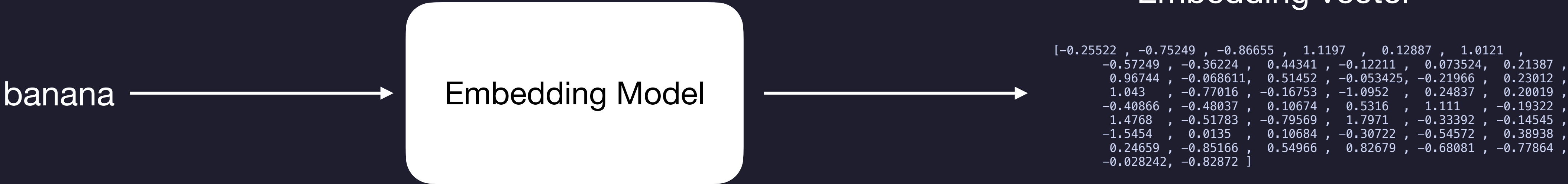
Tabular data → Natural vectorial representation

What about Images, Text, Audio?

Neural networks - CNN, MLP, Transformer architectures

“Representation learning”

Embedding Vectors and Semantic Similarity



Between grape and aeroplane, which one should be closer to banana?

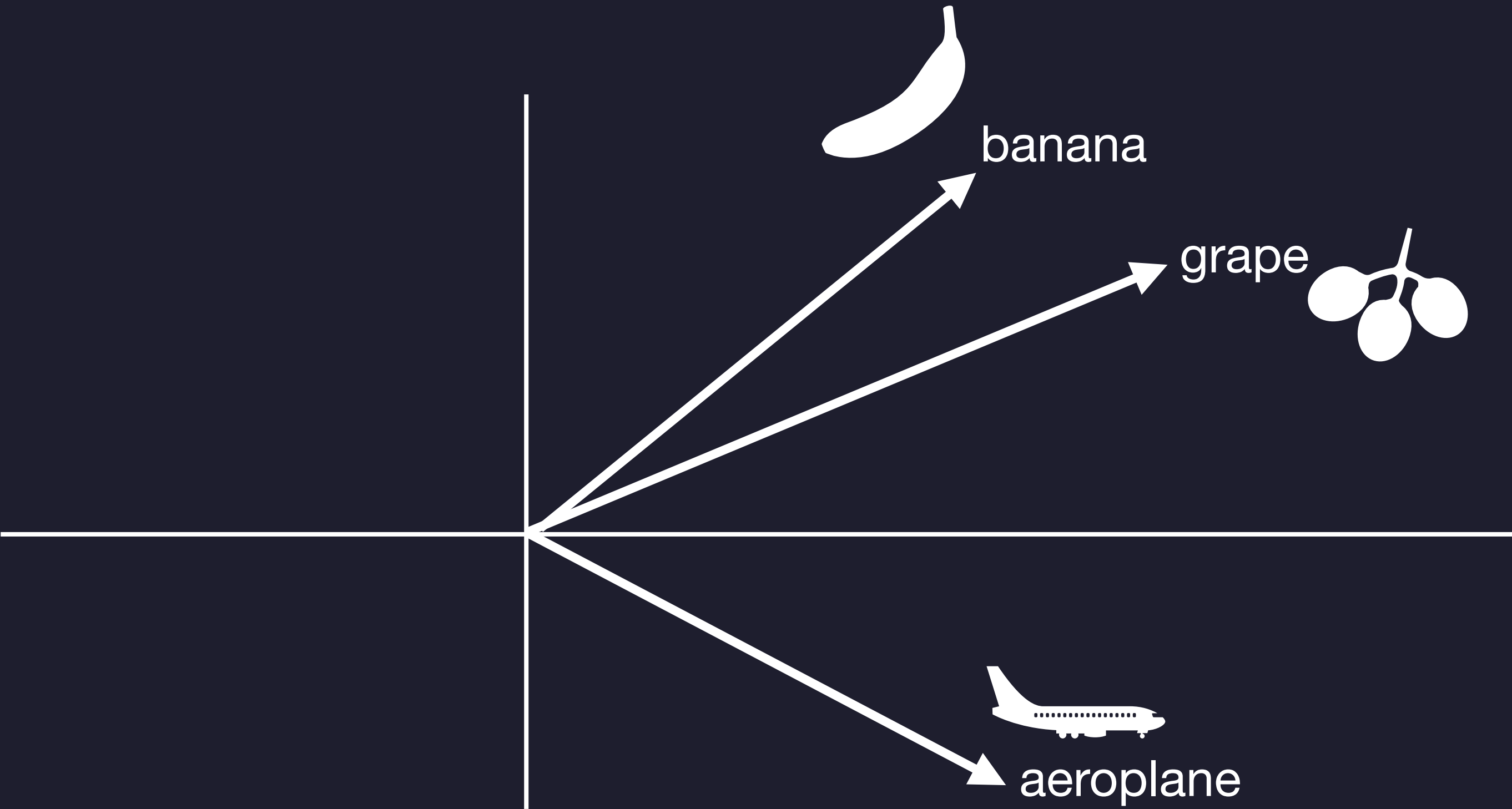
Why should grape be closer?

Distributional Hypothesis

Words that occur in similar contexts tend to have similar meanings

“A word is known by the company it keeps”

Can we validate this numerically?



Document Similarity Search and Reranking

Documents (one per line)

aeroplane

grape

Query

banana

K (number of results)

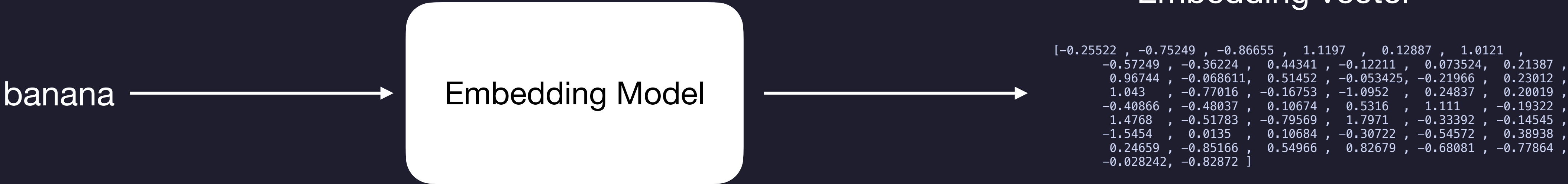
2

Retrieve Top K

Top-K Results

Document	Score
grape	0.47821459174156...
aeroplane	0.23929074406623...

Embedding Vectors and Semantic Similarity



Between grape and aeroplane, which one should be closer to banana?

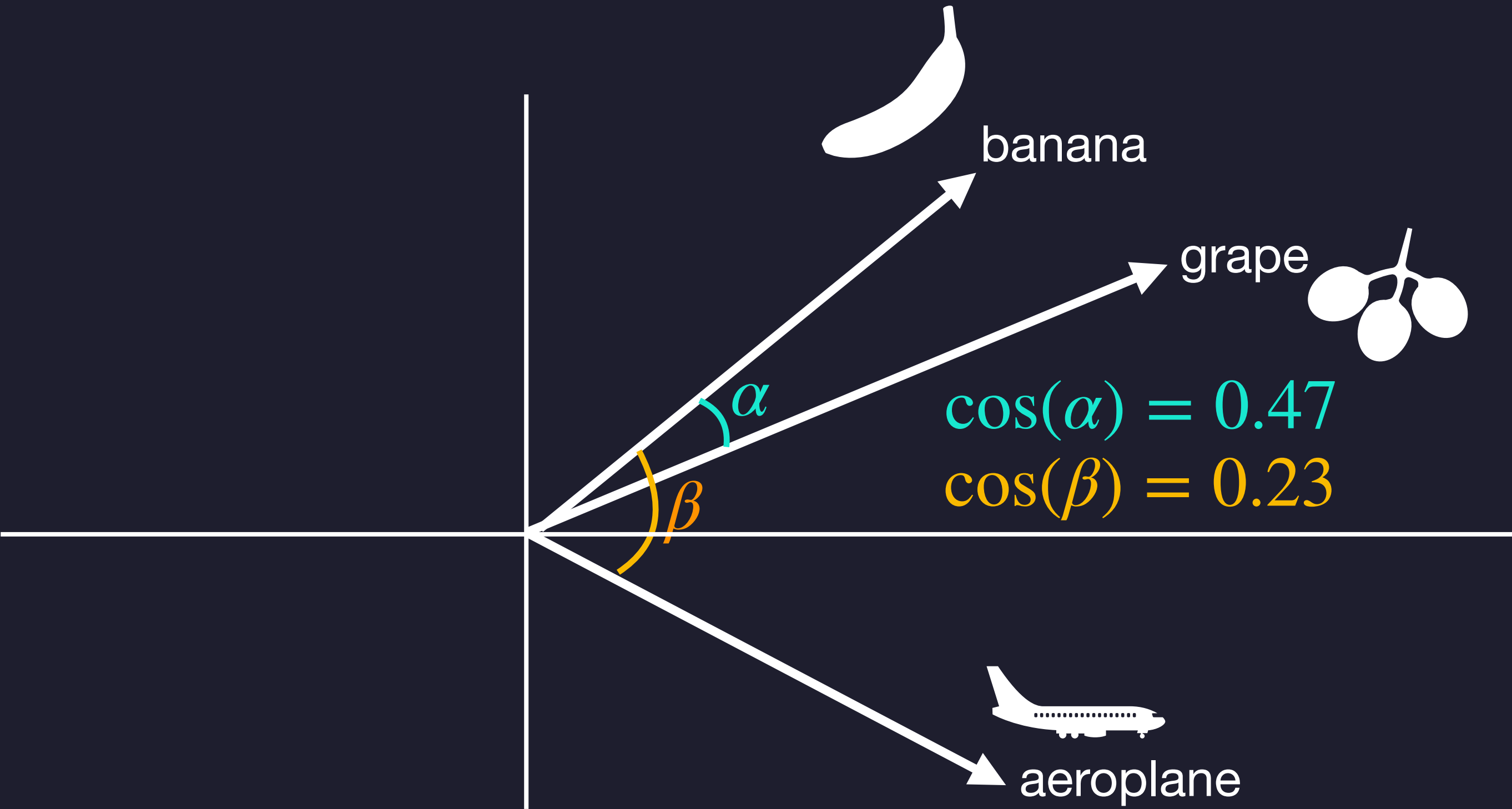
Why should grape be closer?

Distributional Hypothesis

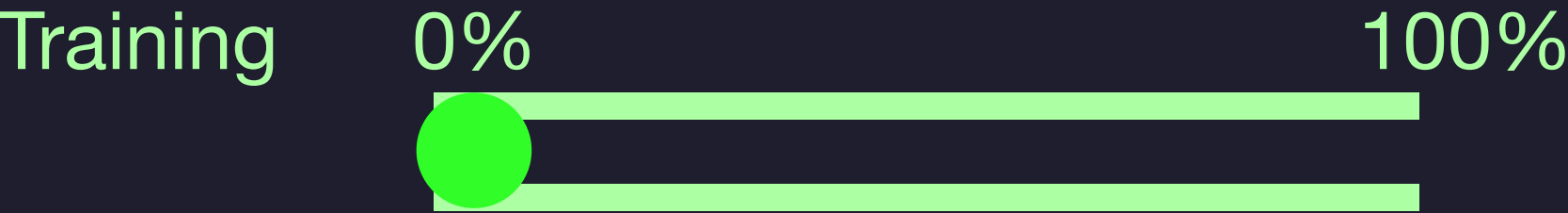
Words that occur in similar contexts tend to have similar meanings

“A word is known by the company it keeps”

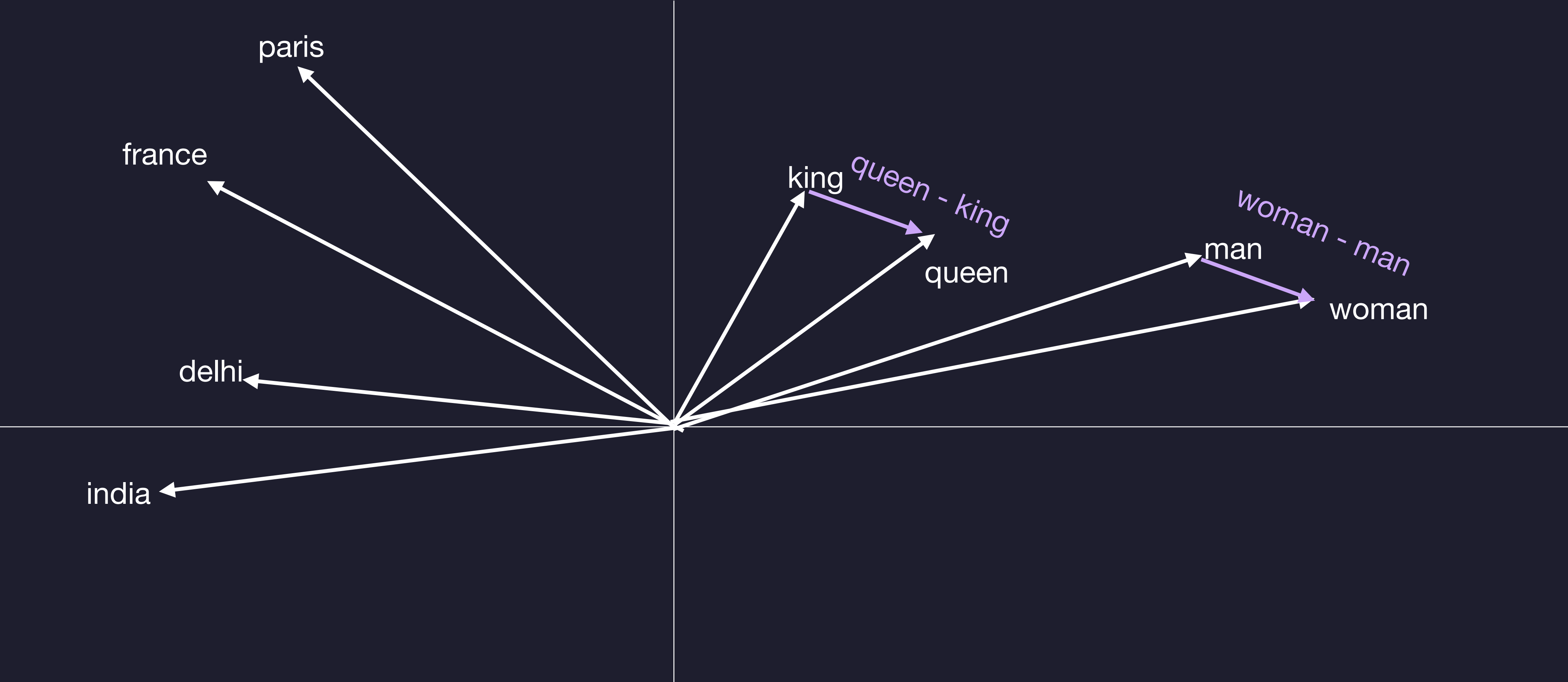
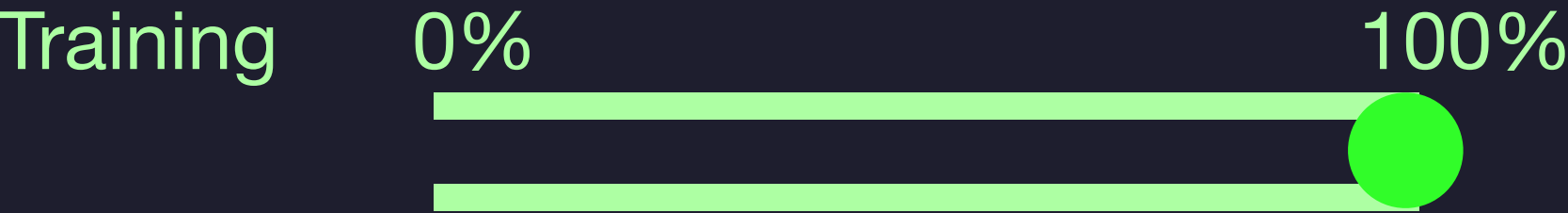
Can we validate this numerically?



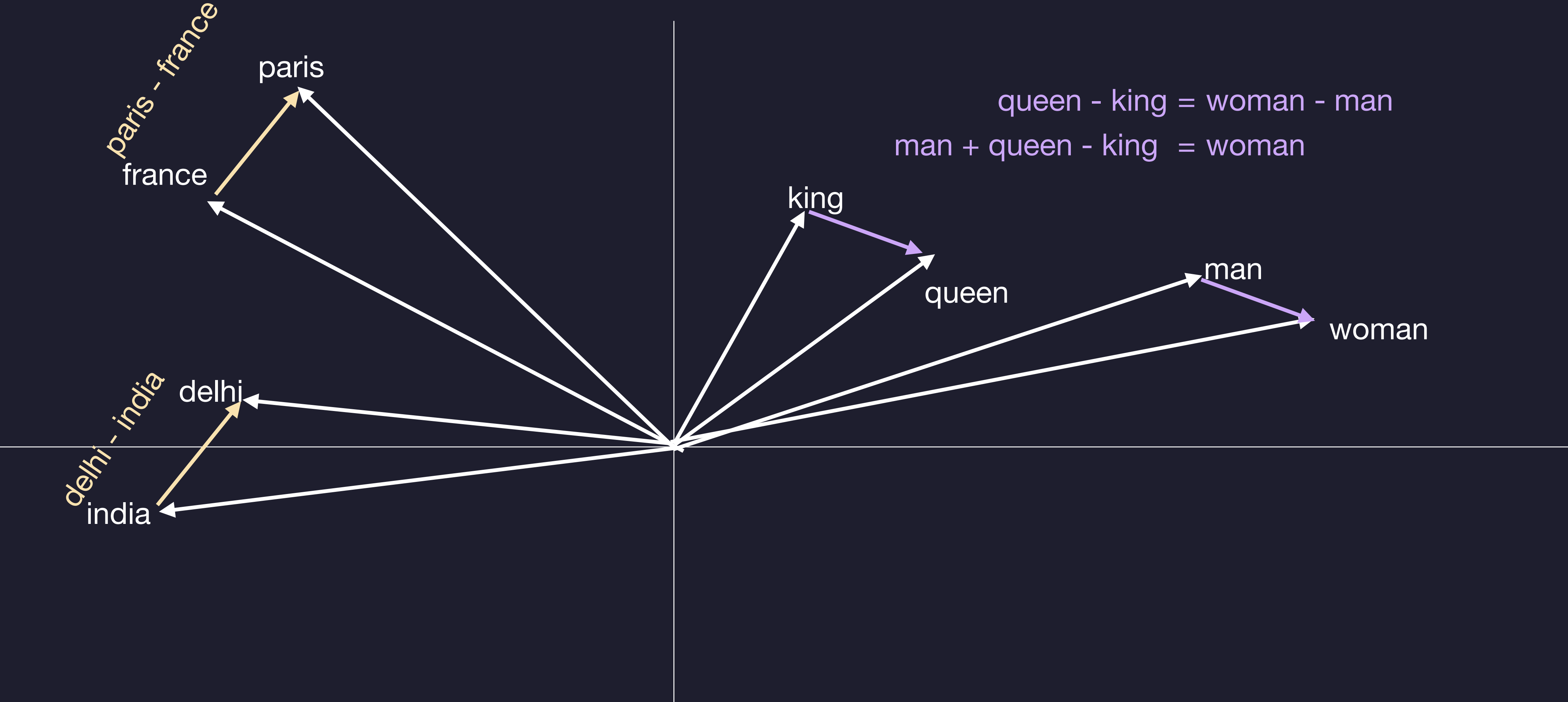
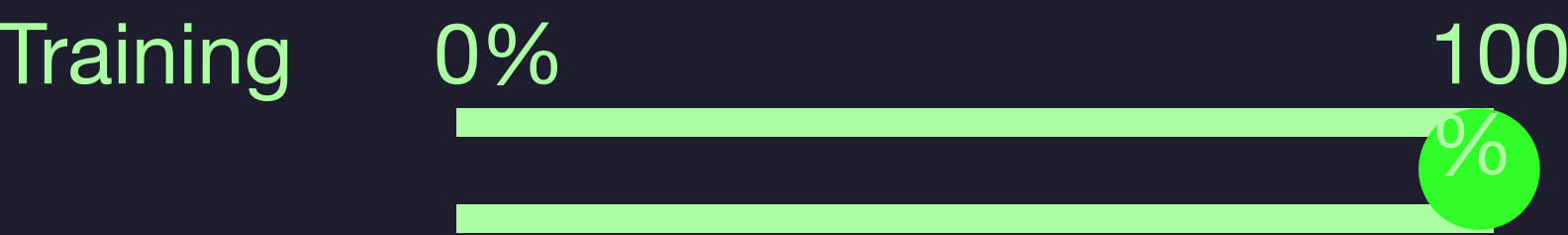
Embedding Vectors



Embedding Vectors



Embedding Vectors



Document Similarity Search and Reranking

Documents (one per line)

COVID-19 is a disease caused by a virus. The most common symptoms are fever, chills, and sore throat, but there are a range of others.

Dementia has the following symptom: Experiencing memory loss, poor judgment, and confusion.

Query

COVID-19 Symptoms

K (number of results)

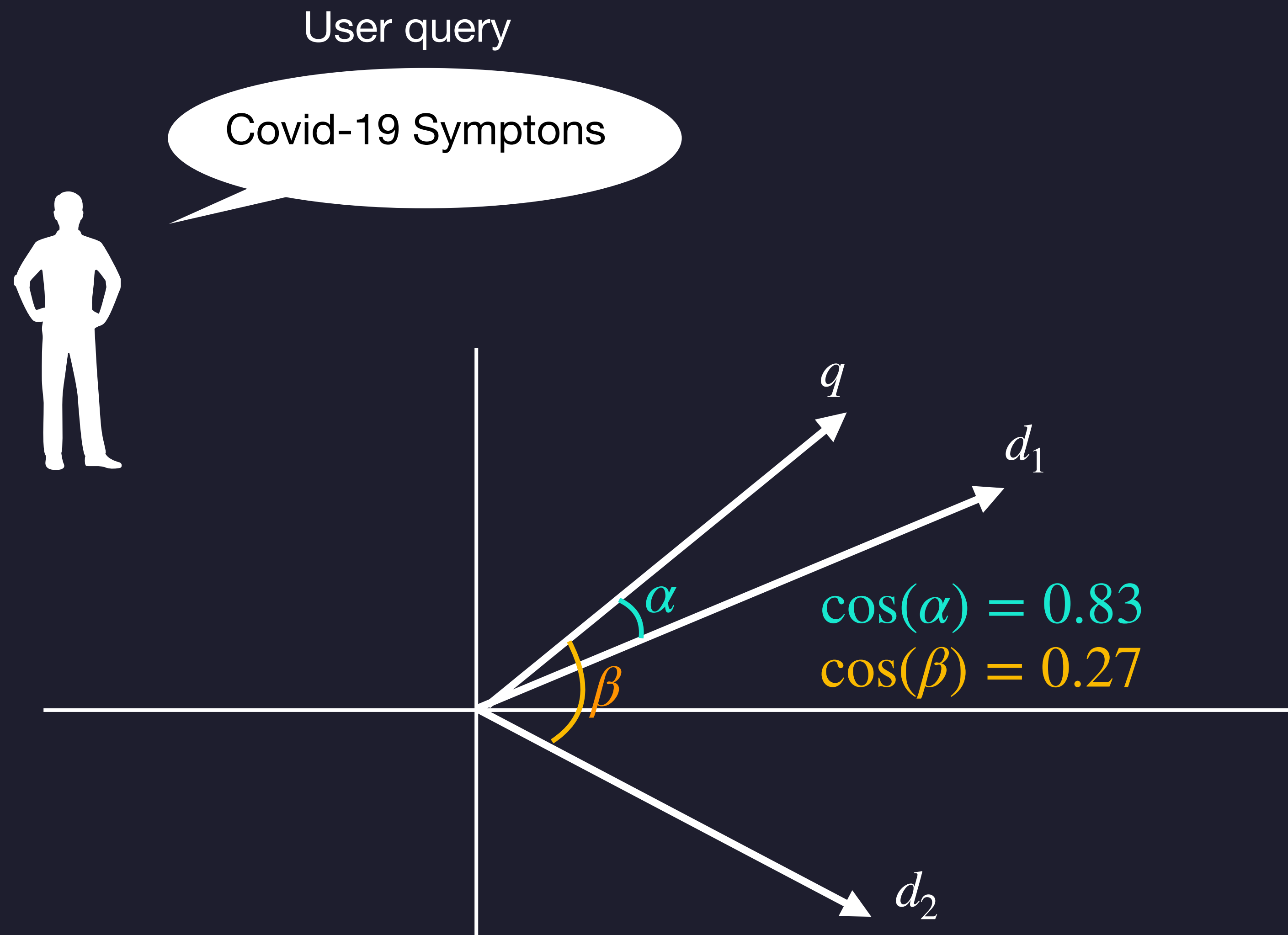
2

Retrieve Top K

Top-K Results

Document	Score
COVID-19 is a disease caused by a virus. The most common symptoms are fever, chills, and sore throat, but there are a range of othe...	0.8346440'
Dementia has the following symptom: Experiencing memory loss, poor judgment, and confusion.	0.2796711'

Embedding Vectors and Semantic Similarity



Doc 1

COVID-19 is a disease caused by a virus. The most common symptoms are fever, chills, and sore throat, but there are a range of others.

Doc 2

Dementia has the following symptom: Experiencing memory loss, poor judgment, and confusion.

Doc 3

COVID-19 has many symptoms

Doc 4

COVID-19 symptoms are bad

Document Similarity Search and Reranking

Documents (one per line)

COVID-19 is a disease caused by a virus. The most common symptoms are fever, chills, and sore throat, but there are a range of others.

Dementia has the following symptom: Experiencing memory loss, poor judgment, and confusion.

COVID-19 has many symptoms.

COVID-19 symptoms are bad.

Query

COVID-19 Symptoms

K (number of results)

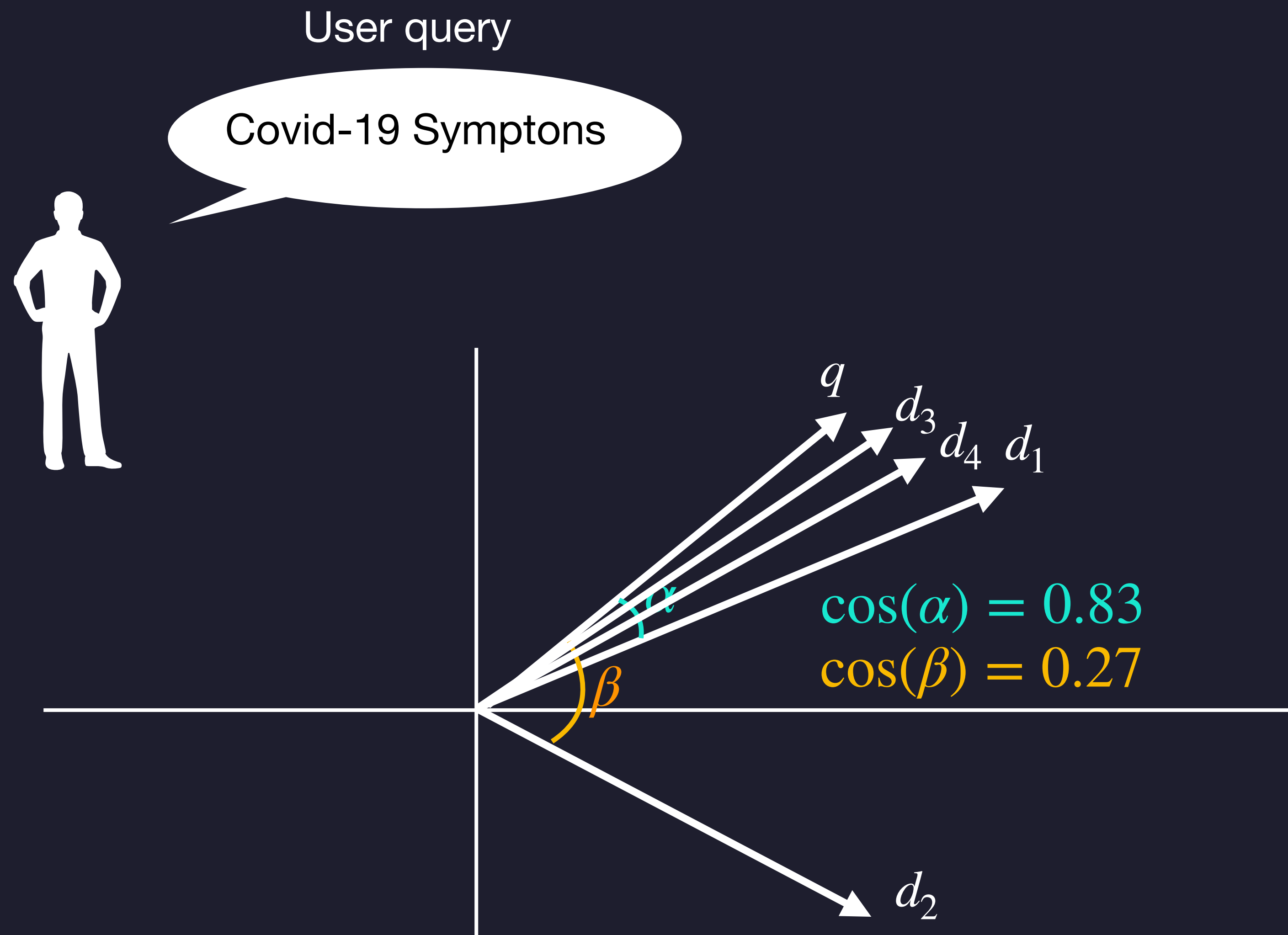
4

Retrieve Top K

Top-K Results

Document	Score
COVID-19 has many symptoms.	0.9215288
COVID-19 symptoms are bad.	0.8622972
COVID-19 is a disease caused by a virus. The most common symptoms are fever, chills, and sore throat, but there are a range of othe...	0.8346440
Dementia has the following symptom: Experiencing memory loss, poor judgment, and confusion.	0.2796711

Embedding Vectors and Semantic Similarity



Doc 1

COVID-19 is a disease caused by a virus. The most common symptoms are fever, chills, and sore throat, but there are a range of others.

Doc 2

Dementia has the following symptom: Experiencing memory loss, poor judgment, and confusion.

Doc 3

COVID-19 has many symptoms

Doc 4

COVID-19 symptoms are bad

Pros and Cons of Vector Similarity - The Need for ReRanking

The good

Fast computation using dot products

- Makes it efficient for large-scale retrieval systems

Naturally handles documents of different lengths

Well-established with extensive library support and vector databases

- Faiss, Chroma, Qdrant, Weaviate

The bad

Ignores term frequency importance

- Treating rare and common words equally

Misses out on richness of document

Long-tail data inaccuracies

- Words that rarely occur in general data may not get the right treatment

Document Similarity Search and Reranking

Documents (one per line)

COVID-19 is a disease caused by a virus. The most common symptoms are fever, chills, and sore throat, but there are a range of others.

Dementia has the following symptom: Experiencing memory loss, poor judgment, and confusion.

COVID-19 has many symptoms.

COVID-19 symptoms are bad.

Query

COVID-19 Symptoms

K (number of results)

4

Retrieve Top K

Top-K Results

Document	Score
COVID-19 has many symptoms.	0.92152885
COVID-19 symptoms are bad.	0.86229725
COVID-19 is a disease caused by a virus. The most common symptoms are fever, chills, and sore throat, but there are a range of othe...	0.83464405
Dementia has the following symptom: Experiencing memory loss, poor judgment, and confusion.	0.27967115

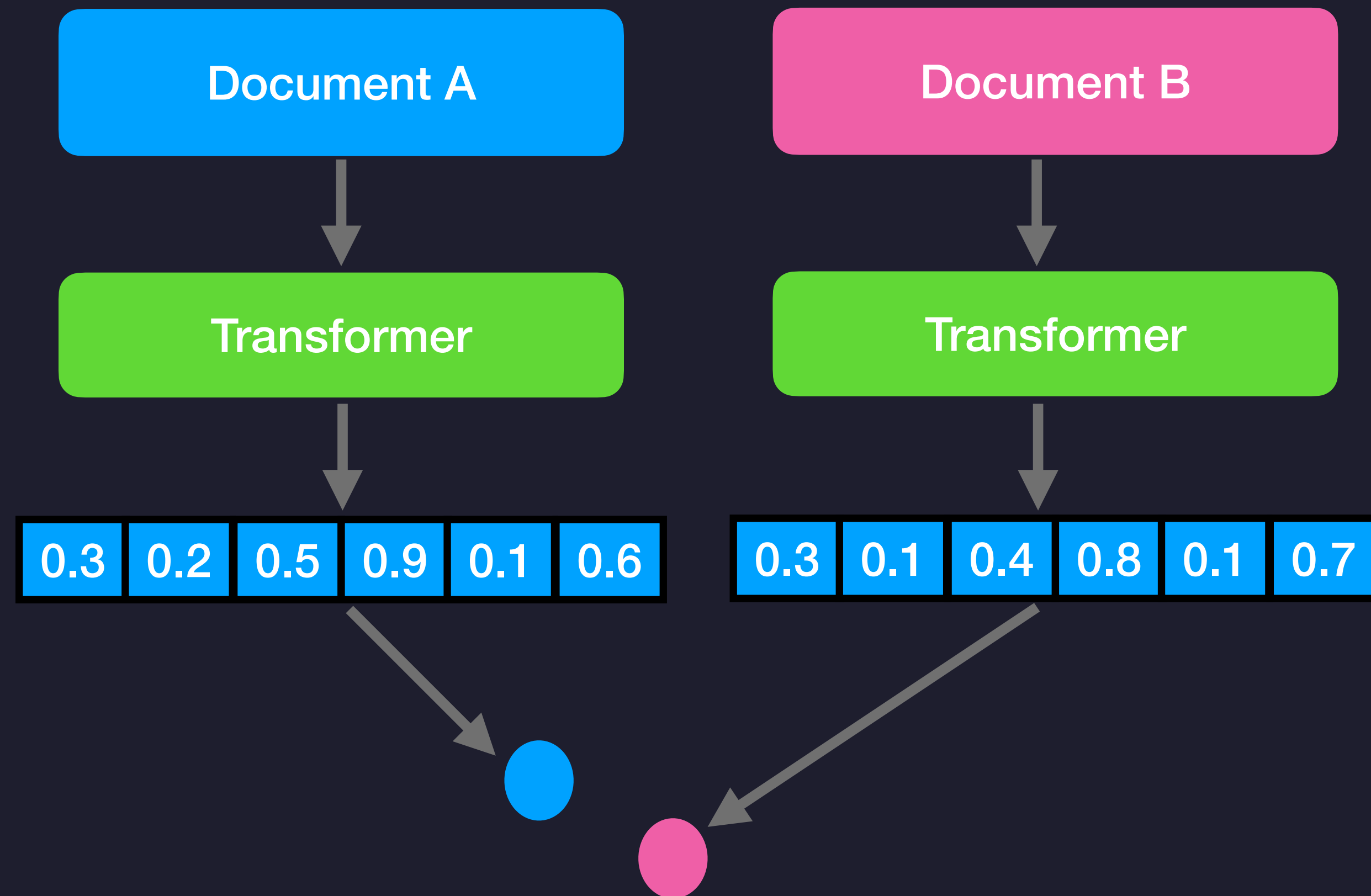
Rerank

Reranked Results

Document	Score
COVID-19 is a disease caused by a virus. The most common symptoms are fever, chills, and sore throat, but there are a range of othe...	9.81164745
COVID-19 symptoms are bad.	7.65879345
COVID-19 has many symptoms.	7.16936345
Dementia has the following symptom: Experiencing memory loss, poor judgment, and confusion.	-9.93092755

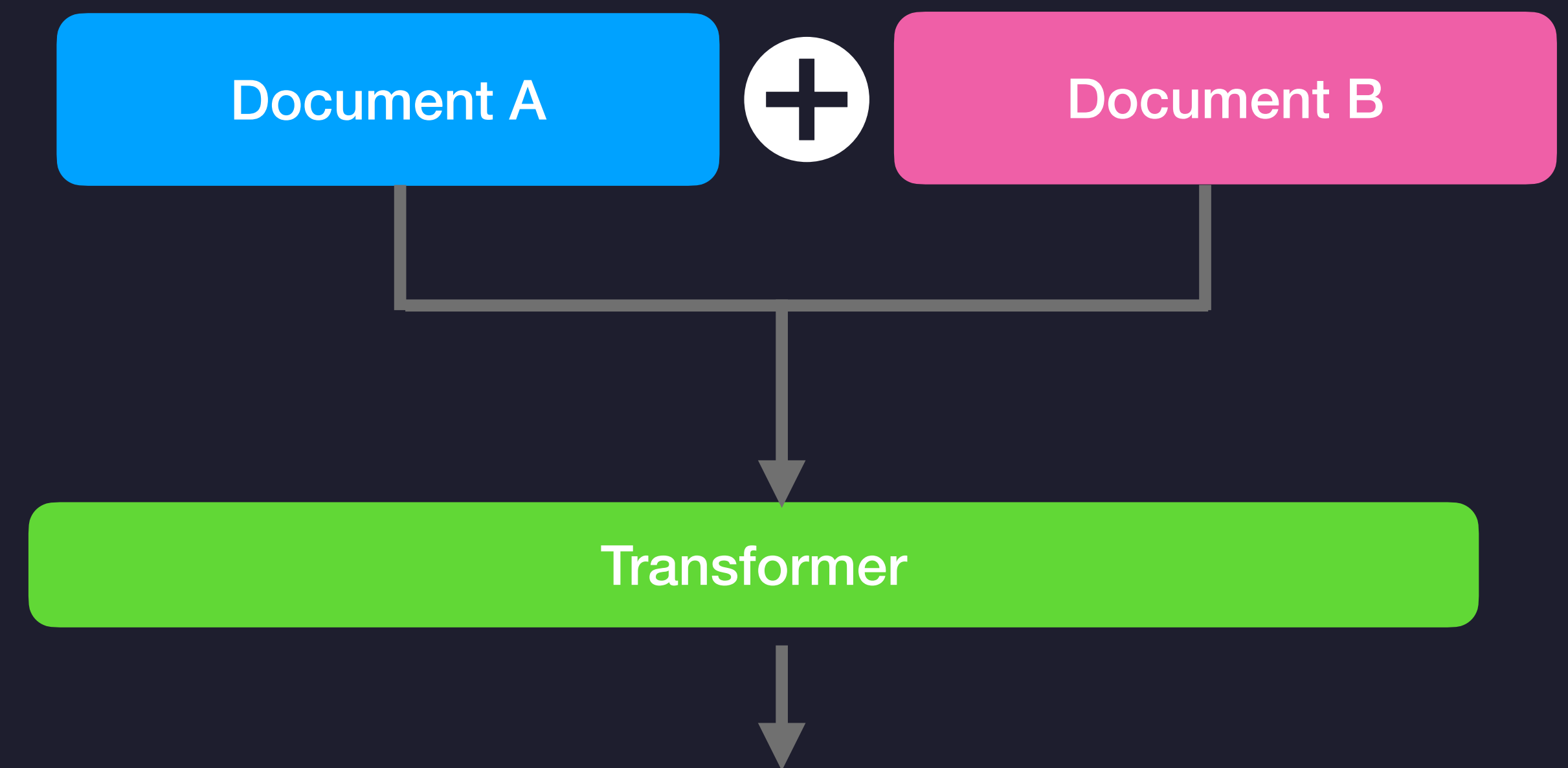
Bi-Encoders and Cross-Encoders

Bi-Encoder



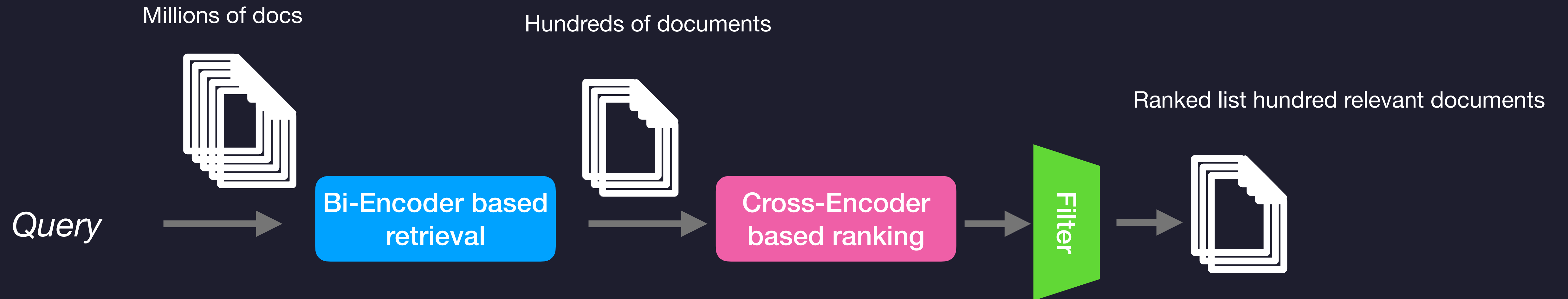
Calculate proximity to get similarity

Cross-Encoder



Similarity Score

Two stage Retrieval



Evaluating the Retrieval

Precision and Recall

Q 1

Doc 1	✓
Doc 2	✓
Doc 3	✗
Doc 4	✗
Doc 5	✗
Doc 6	✓

K = 2

$\text{Prec@2} = 2/2$
 $\text{Recall@2} = 2/3$

K = 5

$\text{Prec@5} = 2/5$
 $\text{Recall@5} = 2/3$

Q 2

Doc 1	✗
Doc 2	✓
Doc 3	✗
Doc 4	✗
Doc 5	✓
Doc 6	✓

$\text{Prec@2} = 1/2$
 $\text{Recall@2} = 1/3$

$\text{Prec@5} = 2/5$
 $\text{Recall@5} = 2/3$

Q 3

Doc 1	✗
Doc 2	✗
Doc 3	✓
Doc 4	✓
Doc 5	✓
Doc 6	✗

$\text{Prec@2} = 0/2$
 $\text{Recall@2} = 0/3$

$\text{Prec@5} = 3/5$
 $\text{Recall@5} = 3/3$

Evaluating the Retrieval

Mean Average Precision

Q 1

Doc 1	✓
Doc 2	✓
Doc 3	✗
Doc 4	✗
Doc 5	✗
Doc 6	✓

Q 2

Doc 1	✗
Doc 2	✓
Doc 3	✗
Doc 4	✗
Doc 5	✓
Doc 6	✓

Q 3

Doc 1	✗
Doc 2	✗
Doc 3	✓
Doc 4	✓
Doc 5	✓
Doc 6	✗

Average Precision

$$\frac{\text{Prec@1} + \text{Prec@2} + \text{Prec@6}}{3}$$

$$\frac{(1/1) + (2/2) + (3/6)}{3} = \frac{2.5}{3}$$

$$\frac{\text{Prec@2} + \text{Prec@5} + \text{Prec@6}}{3}$$

$$\frac{(1/2) + (2/5) + (3/6)}{3} = \frac{1.4}{3}$$

$$\frac{\text{Prec@3} + \text{Prec@4} + \text{Prec@5}}{3}$$

$$\frac{(1/3) + (2/4) + (3/5)}{3} = \frac{1.43}{3}$$

Mean Average Precision

$$\frac{1}{3} \left(\frac{2.5}{3} + \frac{1.4}{3} + \frac{1.43}{3} \right) = 0.59$$

Evaluating the Retrieval

Mean Reciprocal Rank

Q 1

Doc 1 
Doc 2 
Doc 3 
Doc 4 
Doc 5 
Doc 6 

Reciprocal Rank

$$RR = \frac{1}{1}$$

Mean Reciprocal Rank

Q 2

Doc 1 
Doc 2 
Doc 3 
Doc 4 
Doc 5 
Doc 6 

$$RR = \frac{1}{2}$$

Q 3

Doc 1 
Doc 2 
Doc 3 
Doc 4 
Doc 5 
Doc 6 

$$RR = \frac{1}{3}$$

$$MRR = \frac{1}{3}(1 + 0.5 + 0.33) = 0.61$$